

RICHMOND OPERATION CENTRE, BC, Canada

WMO: 711120

Lat: 49.182N

Lon: 123.078W

Elev: 16

StdP: 101.13

Time Zone: -8.00 (NAP)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.6	0.1	-9.8	1.6	0.9	-6.2	2.2	1.4	17.0	7.8	14.6	6.8	2.1	70	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.6	18.9	15.2	17.5	14.8	16.6	14.4	15.7	17.8	15.1	16.9	14.7	16.2	3.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	10.5	16.4	14.2	10.1	15.8	13.9	9.9	15.4	43.9	18.1	42.2	16.9	40.9	16.3	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.0	10.2	DB	-3.5	22.9	2.7	1.1	-5.4	23.7	-7.0	24.3	-8.5	25.0	-10.4	25.8	(4)
(5)			WB	-5.3	17.4	2.8	1.2	-7.3	18.3	-9.0	19.0	-10.5	19.7	-12.6	20.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	5.8	5.9	7.0	8.5	10.6	12.3	13.7	14.2	13.2	10.5	7.7	5.9	(6)	
	DBStd	3.66	2.36	2.95	1.90	1.76	1.76	1.52	1.35	1.29	1.74	1.73	2.09	2.56	(7)	
	HDD10.0	619	130	115	94	51	12	1	0	0	1	14	73	128	(8)	
	HDD18.3	3176	388	347	352	296	240	180	143	128	153	243	320	386	(9)	
	CDD10.0	487	1	1	1	6	31	71	115	131	97	30	3	1	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	4.1	5.0	4.4	4.5	4.3	4.2	4.2	3.4	3.1	2.8	3.6	4.9	4.9	(14)	
(15) Precipitation	PrecAvg	1209	159	126	113	76	65	48	39	41	63	121	174	185	(15)	
	PrecMax	1518	274	286	197	145	147	138	87	106	138	295	358	313	(16)	
	PrecMin	843	28	34	48	16	23	14	0	0	0	21	49	56	(17)	
	PrecStd	185	61	60	34	34	30	30	27	29	35	64	67	56	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	12.6	12.8	15.6	17.5	20.5	20.7	20.2	20.7	16.5	12.6	11.3	(19)	
		MCWB	8.4	9.4	9.1	10.3	13.6	14.1	16.1	16.3	15.4	13.2	11.0	9.3	(20)	
	2%	DB	10.0	11.1	11.4	13.6	15.8	16.8	17.7	18.3	18.7	14.8	12.0	10.2	(21)	
		MCWB	8.9	8.6	8.8	10.0	12.9	14.1	15.1	15.6	14.7	12.3	11.1	9.4	(22)	
	5%	DB	9.2	10.1	10.5	12.2	14.3	15.6	16.7	17.2	17.0	13.8	11.2	9.9	(23)	
		MCWB	8.4	8.6	8.5	9.9	12.2	13.3	14.5	15.0	14.4	12.0	10.5	9.3	(24)	
	10%	DB	8.9	9.2	9.9	11.2	13.4	14.6	15.9	16.4	15.8	12.9	10.8	9.2	(25)	
		MCWB	8.2	8.3	8.4	9.4	11.4	12.8	14.1	14.6	14.0	11.7	10.1	8.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	10.5	10.5	11.8	14.5	15.2	16.4	16.7	16.2	13.9	12.0	10.3	(27)	
		MCDB	10.1	11.3	11.1	13.2	16.4	17.9	19.5	19.3	19.1	15.3	12.1	10.6	(28)	
	2%	WB	9.3	9.7	9.9	11.0	13.4	14.3	15.3	15.8	15.2	13.1	11.5	9.9	(29)	
		MCDB	9.4	10.1	10.4	12.6	15.2	16.2	17.3	17.6	17.3	14.0	11.7	10.0	(30)	
	5%	WB	8.8	9.1	9.2	10.4	12.6	13.6	14.7	15.3	14.7	12.6	10.9	9.3	(31)	
		MCDB	9.0	9.5	9.9	11.6	14.1	15.2	16.3	16.8	16.4	13.3	11.3	9.5	(32)	
	10%	WB	8.2	8.5	8.7	9.8	11.7	13.1	14.2	14.8	14.1	12.0	10.2	8.8	(33)	
		MCDB	8.5	9.0	9.5	10.9	13.1	14.3	15.6	16.1	15.6	12.7	10.6	9.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	4.2	4.2	4.2	4.3	3.7	3.8	3.6	3.9	3.6	3.6	3.4	(35)	
		MCDBR	3.2	4.5	5.3	5.7	5.4	5.1	4.9	5.1	5.9	4.8	3.4	3.0	(36)	
	5% WB	MCWBR	2.9	3.1	4.0	4.0	3.7	3.5	3.1	3.3	3.6	3.5	3.4	2.8	(37)	
		MCWBR	2.7	3.4	3.6	4.4	4.9	4.5	4.7	4.5	5.1	3.8	3.0	2.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.320	0.314	0.331	0.364	0.390	0.364	0.362	0.365	0.349	0.338	0.322	0.309	(40)	
		taud	2.437	2.500	2.474	2.388	2.347	2.442	2.474	2.478	2.526	2.539	2.498	2.450	(41)	
	Ebn at Noon	Ebn at Noon	735	844	885	884	870	895	892	873	856	804	730	698	(42)	
		Edh at Noon	64	75	91	109	119	109	104	99	86	72	60	56	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.86	1.72	2.58	4.02	5.25	5.64	6.30	5.34	3.70	2.03	1.01	0.69	(44)	
	RadStd	RadStd	0.14	0.26	0.35	0.43	0.45	0.67	0.55	0.42	0.33	0.28	0.19	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (69 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page